

## AIR WAR AT SEA

IN THE EARLY YEARS OF AVIATION, navies were, on balance, more aware of the potential of aircraft than were armies. This was especially true of Britain's Royal Navy, where the influence of an imaginative, progressive-minded First Lord of the Admiralty, Winston Churchill, made itself felt. But when the war started, there was no effective way of taking aircraft to sea with the fleet. In September 1914, the Royal Navy converted three cross-Channel steamers into seaplane carriers. The seaplanes were winched off the ship to take off from the sea, and lifted back on board after their mission. It sounded simple and effective, but it was not. The seaplanes found taking off and landing at sea impossible, except under highly favorable conditions – they needed exactly the right degree of swell.

Before the war, both the US and British navies had experimented with launching aircraft off a platform on the deck of a ship. The Royal Navy resumed these experiments in earnest in 1917. A light battlecruiser, HMS *Furious*, had its forward guns removed and replaced by a takeoff deck. The idea was that the latest land-based aircraft, superior in performance to seaplanes, would take off and land from the ship.

The maneuver was undertaken by Squadron Commander E.H. Dunning flying the highly agile Sopwith Pup biplane. Taking off was relatively easy if the ship sailed into the wind, but landing was another matter. Dunning managed it twice by matching his speed to that of the ship so he



### PLATFORM LAUNCH

*One system for launching aircraft at sea was to put them on a lighter – a sort of barge – towed behind a light, speedy vessel (left). When traveling fast enough, this would generate the windspeed, and hence the lift, needed for takeoff.*



### HOISTED OVERBOARD

*Seaplanes offered a way of carrying aircraft with a fleet. Machines like this Short seaplane were winched overboard from a ship, took off from the sea, and might with luck land safely alongside and be lifted back on board.*



could effectively hover down like a helicopter, helped by other pilots on deck who grabbed rope toggles on the wings and pulled him down as he cut his engine. But his third attempt proved fatal. The aircraft stalled and cartwheeled overboard after a tire blew, and the unfortunate Dunning drowned. *Furious* was then equipped with a rear deck for landing, but the turbulence created by the funnel and central superstructure of the ship tossed the aircraft around too much. Nevertheless, Sopwith Camels launched from *Furious* carried out the first-ever successful air-strike by carrier-borne aircraft, bombing zeppelin sheds at Tondern in July 1918. In the last months